Reconciliation Exception Classification Review

Prepared for the reconciliation-platform workspace. Review date: April 5, 2026. Scope: end-to-end trace from workbook parsing through normalization, pairwise matching, residual exception classification, summary generation, workbook export, API delivery, and web reporting.

VERIFIED FINDING INFERRED RECOMMENDATION Verified findings are grounded in the current source, tests, docs, and surfaced outputs. Recommendations are limited to changes supported by the implementation already present.

Executive Summary

The project is already capable of actionable exception classification. It no longer relies on a user-facing generic unmatched bucket. Instead, it parses workbook rows, normalizes fields, performs pairwise match evaluation, promotes residual cases into typed exception categories, builds summary metrics from the final exception set, exports those categories to workbook sheets, exposes them over API, and renders them in the web UI.

The largest remaining gap is architectural rather than taxonomic. Grouped categories such as `LikelySplitCombineIssue` and `DepositGroupingIssue` are implemented only as a bounded residual stage after greedy one-to-one matching has already consumed candidates. The second-largest gap is audit lineage: promoted classifications such as Date Variance to Timing Difference or Likely Miscoding are explainable, but the full original-to-final decision chain is not preserved.

> **Bottom line:** the platform already supports a meaningful, accountant-facing exception taxonomy end-to-end. The next step is to strengthen grouped reasoning and preserve richer rule lineage, not to invent new category names.

Current-State Findings

1. VERIFIED **High:** The runtime path is explicit and connected from `packages/core/src/parsers/excelParser.ts` through `packages/core/src/normalization/transactions.ts`, `packages/core/src/matching/engine.ts`, `packages/core/src/reporting/exceptionClassifier.ts`, `packages/core/src/reporting/summary.ts`, `packages/core/src/export/workbookExporter.ts`, `apps/api/src/app.ts`, and `apps/web/src/components/ExceptionTable.tsx`.
2. VERIFIED **Medium:** The surfaced reporting model is category-specific. Shared taxonomy is defined in `packages/domain/src/enums.ts`, display names and next actions in `packages/domain/src/messages.ts`, and those categories reach both workbook and web output.
3. VERIFIED **High:** Grouped classification is partial because grouped reasoning is residual-only and capped to three grouped members. Oversized grouped scenarios are already proven to fall back to missing-item categories.
4. VERIFIED **Medium:** Promotion logic from Date Variance to Timing Difference or Likely Miscoding is useful, but it overwrites the original classification path instead of preserving both traces.
5. VERIFIED **Medium:** Grouped exceptions preserve rich evidence text, but only one structured counterpart is stored, which limits reviewer traceability for multi-row issues.

6. VERIFIED **Low:** Internal fallback terms such as `Unmatched` and defensive handling for unknown summary categories still exist, but they are not the user-facing reporting model.

Classification Coverage Matrix

Classification	Supported?	Evidence in Code	Gaps	Recommendation
timing difference	Yes	Extended-window candidates come from the matcher and residual timing is emitted by the classifier before missing fallback.	Promotion from Date Variance does not preserve lineage.	Keep original and promoted traces together.
missing in bank	Yes	Unmatched QBO rows fall through to Missing from Bank after stronger stages are exhausted.	Depends on earlier residual coverage being complete enough.	Keep as a final fallback only.
missing in QBO	Yes	Unmatched bank rows fall through to Missing from QBO after stronger stages are exhausted.	Depends on earlier residual coverage being complete enough.	Keep as a final fallback only.
duplicate in bank	Yes	Same-source duplicate detection runs before cross-source matching and yields a bank-specific category.	No duplicate subtype beyond the main category.	Accept current design unless reversal-specific handling becomes necessary.
duplicate in QBO	Yes	Same-source duplicate detection yields a QBO-specific category.	No duplicate subtype beyond the main category.	Accept current design unless reversal-specific handling becomes necessary.
amount variance	Yes	The implemented category is Amount Mismatch, emitted by the matcher and surfaced as a final exception.	The prompt term and code term differ.	Document Amount Mismatch as the canonical implementation of amount variance.
date variance	Yes	Extended-window same-amount matches are classified as Date Variance unless stronger evidence promotes them.	Some Date Variance cases are promoted away.	Preserve promotion lineage.
likely miscoding	Yes	Residual same-amount near-date conflicts and some Date Variance pairs become Likely Miscoding.	Depends on text and reference conflict heuristics.	Keep thresholds deterministic and configurable where practical.

Classification	Supported?	Evidence in Code	Gaps	Recommendation
likely split/combine issue	Partial	Grouped residual logic emits this category for balanced one-to-many and many-to-one residuals.	Residual-only and capped to three grouped members.	Promote grouped reasoning into a staged matcher with configurable bounds.
stale outstanding check	Yes	Unmatched aged QBO check-like disbursements are promoted before Missing from Bank fallback.	The threshold is fixed at 45 days.	Make stale thresholds configurable by account or source profile.
deposit grouping issue	Partial	Grouped positive residuals with deposit-like kinds are emitted as Deposit Grouping Issue.	Residual-only and capped to three grouped members.	Move grouped deposit logic into a first-class residual stage with full member traceability.

Logic and Architecture Assessment

The current design is already materially stronger than a generic unmatched-reporting workflow. The taxonomy is explicit, shared, and surfaced consistently. Summary metrics are built from the final exception set instead of a separate hand-maintained reporting layer, which reduces drift between details and totals.

The main architectural weakness is that the matching engine is still fundamentally greedy and pairwise. That is acceptable for exact matches, reference variance, description variance, and amount mismatch, but it means grouped categories are a downstream repair step rather than a native matching capability.

Reporting and Auditability Assessment

Reporting is already actionable for accounting staff. Exception items carry category, severity, rule trace, evidence, recommended next actions, and counterpart references where they exist. The workbook Error Report and the web exception table both preserve those fields.

Auditability is good but not complete. The biggest reporting limitation is grouped traceability: grouped categories preserve evidence text, but not a full structured counterpart set. The second limitation is rule-lineage visibility for promoted exceptions.

Prioritized Recommendations

Immediate Improvements

Title: Preserve lineage for promoted classifications

Problem observed: Date Variance can be promoted into Timing Difference or Likely Miscoding, but the original classification path is not retained.

Why it matters: Reviewers can see the final category, but not the original decision that was superseded.

Severity: Medium

Implementation complexity: Medium

Recommended solution: Extend the rule trace so promoted exceptions retain both the originating match trace and the final classification trace.

Impacted files/modules: `packages/core/src/reporting/exceptionClassifier.ts`, `packages/domain/src/types.ts`, `apps/web/src/components/ExceptionTable.tsx`

Change type: logic, data model, UI-reporting, test coverage

Suggested acceptance criteria: Promoted exceptions display both original and final traces in API output, workbook export, and UI rendering.

Title: Expose full grouped counterpart sets structurally

Problem observed: Grouped categories keep only one structured counterpart even when the issue is inherently multi-row.

Why it matters: Grouped exception review needs explicit member-level traceability, not just concatenated evidence text.

Severity: Medium

Implementation complexity: Medium

Recommended solution: Add structured counterpart arrays for grouped categories and render the full member set in workbook and web reports.

Impacted files/modules: `packages/core/src/reporting/exceptionClassifier.ts`, `packages/domain/src/types.ts`, `packages/core/src/export/workbookExporter.ts`, `apps/web/src/components/ExceptionTable.tsx`

Change type: data model, UI-reporting, documentation, test coverage

Suggested acceptance criteria: Grouped exceptions render all linked counterpart file and line references without relying on evidence text alone.

Near-Term Enhancements

Title: Make grouped and stale-check policy configurable

Problem observed: Grouped search depth and stale-check age are hard-coded.

Why it matters: These are reconciliation-policy decisions and may vary by account, source, or business cycle.

Severity: Medium

Implementation complexity: Low

Recommended solution: Add config for stale outstanding check days, grouped candidate pool size, and grouped maximum match size.

Impacted files/modules: `packages/core/src/reporting/exceptionClassifier.ts`, `packages/config/src/default-config.ts`, `packages/domain/src/types.ts`, `docs/reconciliation-rules.md`

Change type: logic, data model, documentation, test coverage

Suggested acceptance criteria: Account overrides can adjust grouped and stale-check policy and the regression suite covers both default and overridden behavior.

Title: Retire the remaining 5703-specific provenance branch from the general classifier flow

Problem observed: The classifier still contains hard-coded 5703 timing provenance logic.

Why it matters: It weakens the rule-driven and generalizable nature of the exception-classification path.

Severity: Medium

Implementation complexity: Medium

Recommended solution: Isolate the logic behind explicit config or replace it with fixture-backed general rules.

Impacted files/modules: `packages/core/src/reporting/summary.ts`, `packages/core/src/reporting/exceptionClassifier.ts`, `packages/core/src/reporting/summary.edge-cases.test.ts`

Change type: architectural, logic, test coverage, documentation

Suggested acceptance criteria: No filename-or-volume heuristic is required in the normal timing provenance path.

Structural Redesign Items

Title: Promote grouped reasoning into a staged residual engine

Problem observed: Grouped categories are a bounded residual fix after greedy one-to-one matching.

Why it matters: This is the main reason split/combine and deposit grouping remain only partially supported.

Severity: High

Implementation complexity: High

Recommended solution: Preserve pairwise exact matching, then add an explicit staged residual engine for stale checks, miscoding, timing, split/combine, deposit grouping, and only then missing-item fallback.

Impacted files/modules: `packages/core/src/matching/engine.ts`, `packages/core/src/reporting/exceptionClassifier.ts`, `packages/domain/src/types.ts`,

`docs/architecture.md`

Change type: architectural, logic, data model, test coverage

Suggested acceptance criteria: Grouped categories are no longer residual-only, search bounds are configurable, and missing-item categories appear only after grouped reasoning is exhausted.

Quick wins: preserve promotion lineage, expose grouped counterpart arrays, add a UI and API contract test that fails on unknown surfaced categories.

Medium-term refactors: externalize grouped and stale-check policy, remove 5703-specific classifier branching, tighten grouped trace rendering.

Strategic redesign items: staged residual engine, grouped-first residual reasoning, stronger separation between pairwise matching and grouped exception policy.

Suggested Rule Model

1. Parse workbook rows and preserve source file, worksheet, and line-number provenance.
2. Normalize dates, signed amounts, tokens, and data-quality issues.
3. Emit Invalid or Missing Data immediately for rows that cannot safely participate in matching.
4. Detect same-source duplicates before any cross-source matching.
5. Run exact and deterministic pairwise review matching: reference variance, description variance, date variance, amount mismatch, ambiguous review.
6. Run residual stale outstanding check logic for aged unmatched QBO check-like disbursements.
7. Run residual likely miscoding logic for distinctive same-amount near-date conflicts.
8. Run residual timing-difference logic for same-amount compatible-sign items outside primary tolerance and inside the extended timing window.
9. Run grouped residual split/combine and deposit grouping logic with configurable search bounds and full structured member traces.
0. Assign Missing from QBO and Missing from Bank only after all stronger residual categories are exhausted.
1. Build summaries, charts, workbook reports, and API payloads strictly from the final exception set while preserving original rule lineage for promoted categories.

Testing Recommendations

- Keep the combined all-exceptions workbook pair as a release gate proving all categories surface through API and summary totals.
- Add tests for promotion lineage so Date Variance to Timing Difference and Date Variance to Likely Miscoding preserve both original and final traces.
- Add workbook and UI tests that assert grouped exceptions render all counterpart members structurally.
- Add configuration tests for grouped search depth, grouped pool size, and stale-check threshold once those settings become configurable.
- Add a contract test that fails if a surfaced category falls outside the shared taxonomy.

Example Improved Output

Source	File / Line	Category	Amount	Reason	Next Action
bank	bank.xlsx / 118	Missing from QBO	-125.50	Bank transaction remained unmatched after exact, review, timing, grouped, and miscoding stages.	Verify whether the transaction was posted to the correct QBO account and period.
qbo	qbo.xlsx / 140	Missing from Bank	-210.00	QBO transaction remained unmatched after exact, review, timing, grouped, and stale-check screening.	Inspect the bank statement for omitted, reversed, or delayed posting.
bank + qbo	bank.xlsx / 71, qbo.xlsx / 74	Timing Difference	-2400.00	Same amount and timing-compatible transaction kinds fell outside primary tolerance but inside the timing window.	Carry as timing support and monitor the next reconciliation cycle.
bank + qbo	bank.xlsx / 85, qbo.xlsx / 91	Likely Miscoding	-500.00	Same amount and near date, but the narrative and reference context conflict materially.	Review whether the transaction was posted to the wrong vendor, customer, or account.
bank + qbo	bank.xlsx / 102, qbo.xlsx / 103-104	Likely Split/Combine Issue	-300.00	One bank line balances to multiple QBO lines, so grouped review fits better than missing fallback.	Document the one-to-many relationship and retain grouped support.
bank + qbo	bank.xlsx / 118, qbo.xlsx / 119-120	Deposit Grouping Issue	3200.00	One bank deposit balances to multiple deposit-like QBO entries.	Tie the deposit to receipt-batch support and sign off on the grouped member set.

Source	File / Line	Category	Amount	Reason	Next Action
qbo	qbo.xlsx / 155	Stale Outstanding Check	-150.00	Unmatched QBO check-like disbursement aged beyond the stale threshold.	Void, reissue, or retain as outstanding based on current check status.

Closing Assessment

This project has already moved beyond generic unmatched reporting. Its shared taxonomy, final exception summaries, workbook export, API payloads, and web reporting together form a meaningful reconciliation exception-classification system. The most important next step is to improve grouped reasoning and audit lineage, not to broaden the taxonomy further.